HEMANT PANCHAL

TECHNICAL ARCHITECT | ENTERPRISE SOLUTION ARCHITECT | PRINCIPAL ENGINEER

Mumbai, India | +91-8108710030 | career.hemantpanchal67@gmail.com | Total Experience: 15+ Years

EXECUTIVE SUMMARY

Accomplished and strategic **Technical Architect & Enterprise Solution Architect with 15+ years** of rich experience leading end-to-end architecture, solution design, and engineering governance for large-scale enterprise applications across **Banking/FinTech, Enterprise HRMS SaaS (900+ clients), Fortune 14 Global Retail (Costco Wholesale), and High-Concurrency Sports Gaming Platforms.**

Skills: Proven expertise in designing and architecting extensible low-code/no-code enterprise builder ecosystems—including Dynamic Form Creation, Workflow Creation & Orchestration Engines, Business Rule Engines (BRE), Document & Report Engines, SLA & Approval Policy Engines, and Multichannel Notification Engines. Recognized for deep technical mastery across modern .NET ecosystems (.NET Core / .NET 8 / .NET 10), scalable Microservices, Distributed Systems, Event-Driven Architecture (Apache Kafka), Cloud (Azure/AWS), Kubernetes containerization, Database Design (PostgreSQL /SQL Server), AI-augmented developer tooling (Model Context Protocol / MCP - JIRA & Figma), Zero-Trust Security, High Availability, and Architecture Governance (ADRs).
Cloud Azure: Azure App Service, IaaS, VM, Azure Storage Service, Azure Event Grid, Azure Service Bus, Event Hub, Azure Data Factory, Azure Databricks, Azure Docker service, Azure AKS cluster
AWS: EC2, S3 Bucket, AWS Lamda, Serverless

KEY SKILLS & TECHNICAL COMPETENCIES

Architecture & Design	Enterprise Architecture, Solution Design, Microservices, Distributed Systems, Event-Driven Architecture (EDA), Domain-Driven Design (DDD), Clean Architecture, SOLID Principles, High Availability (HA), Scalability, Disaster Recovery (DR), Fault Tolerance, Architecture Decision Records (ADRs), Architecture Governance
Enterprise Builders & Engines	Dynamic Form Creation & Form Builder, Workflow Creation & Orchestration Engine, Dynamic Attribute Builder, Business Rule Engine (BRE), Report Builder & Report Engine, Document Builder & Document Engine, SLA & Approval Policy Builder & Engine, Notification Builder & Event Engine
Languages & Frameworks	.NET Core, .NET 8, .NET 10, ASP.NET Core Web API, ASP.NET MVC, C#, React, Node.js, JavaScript, jQuery, HTML5, CSS3, Windows Services
Databases & Caching	PostgreSQL, Microsoft SQL Server, Oracle, Redis (Distributed Caching), Polyglot Persistence, Schema Design, Performance Indexing & Query Tuning
Messaging & Integration	Apache Kafka, RabbitMQ, Message Brokers, Asynchronous Queuing, RESTful API Architecture, Webhooks, Idempotency, Circuit Breaker, and Resiliency Patterns
Cloud, Containers & DevOps	Microsoft Azure, Amazon Web Services (AWS), Kubernetes, Docker Containerization, CI/CD Automation (GitHub Actions, Azure DevOps), SonarQube, Infrastructure as Code
Security, Performance & QA	Zero-Trust Security, OAuth2, Secrets Management, Audit Logging, JMeter, BlazeMeter, SonarQube Quality Gates
Leadership & Governance	Technical Leadership, Architecture Reviews, Code Quality Enforcement, Tech Debt Remediation, Engineering Mentorship, AI-Assisted Engineering (Model Context Protocol / MCP - JIRA & Figma), Agile/Scrum, Stakeholder Management

WORK EXPERIENCE | INDUSTRY EXPERTISE

Technical Architect | **Enablistar Pvt. Ltd.** **April 2024 – Present**

Domain: Banking & Enterprise Financial Platforms

- **Enterprise-Level Architecture & Development (.NET Core):** Directed end-to-end technical architecture, solution design, and enterprise-level development for mission-critical banking and financial platforms. Engineered highly scalable, resilient cloud-native architectures using .NET Core / .NET 8 microservices, ensuring strict security, high availability, and operational reliability throughout the application lifecycle.
- **Core Low-Code Builder & Engine Ecosystem:** Architected and delivered full-lifecycle enterprise builders and runtime engines:
- Architected an enterprise-level schema-driven Dynamic Form Creation platform and dynamic attribute builder featuring complex multi-level validations, contextual conditional rendering, custom data bindings, and dynamic UI state management.
- Architected and delivered enterprise platform engines for Workflow Orchestration, Business Rules, Reporting, Document Generation, SLA & Approvals, and Omnichannel Notifications, enabling configurable, scalable, and automation-driven business processes.
- Designed state-machine workflow orchestration, declarative Business Rule Engine, self-service Report/Engine, dynamic Document Builder/Engine, hierarchical Approval & SLA Engine, and resilient Notification Engine supporting asynchronous processing, human-in-the-loop workflows, integrations, and distributed execution.
- **Scalable Microservices Architecture (.NET Core):** Architected and implemented distributed, scalable Microservices utilizing .NET Core / .NET 8; defined granular service boundaries, Domain-Driven Design (DDD), Clean Architecture, and resilient RESTful APIs designed for high concurrency and enterprise scale.
- **Event-Driven Streaming & Decoupling (Kafka):** Architected and deployed a resilient, scalable event-driven messaging backbone using Apache Kafka for real-time pub/sub messaging, transaction event streaming, asynchronous inter-service workflows, and decoupled integrations with guaranteed message delivery and idempotency.
- **AI-Augmented Engineering & MCP Integrations:** Pioneered practical AI tool adoption and Model Context Protocol (MCP) integrations across engineering workflows to maximize developer velocity (from a usage and productivity enablement perspective): connected JIRA MCP to automatically decompose architectural specs and generate standardized user stories, tasks, and acceptance criteria; integrated Figma MCP to streamline design-to-code pipelines, directly transforming Figma UI designs into production-ready React UI component code.
- **Technology Stack:** .NET Core, .NET 8, C#, Scalable Microservices, Apache Kafka, PostgreSQL, Redis, React, Model Context Protocol (MCP - JIRA & Figma), Docker, Kubernetes, Azure.
- Led technology strategy and architecture decisions, evaluating frameworks, databases, cloud services, messaging, and caching platforms through technical and cost-benefit analysis.
- Established Clean Architecture, DDD, SOLID, coding standards, API governance, and SonarQube quality gates to drive engineering excellence and maintainability.
- Architected secure, resilient, and highly available APIs and distributed systems, covering OAuth2, authorization, versioning, idempotency, retry, timeouts, backward compatibility, HA/DR, scalability, and business continuity.
- Defined cloud-native infrastructure and DevOps standards including containerization, Kubernetes, CI/CD, automated testing, deployment/rollback, secrets management, observability, and immutable infrastructure.
- Established enterprise security and compliance practices including encryption, Zero Trust principles, centralized secrets, audit logging, and secure data handling.
- Drove architecture governance through ADRs, design reviews, POCs, technology evaluations, and architecture standards aligned with business objectives.
- Provided technical leadership and mentorship, resolving complex architectural and production challenges while guiding engineering teams on scalable and maintainable solutions.

Project Manager / Technical Lead | **Cnergyis Infotech India Pvt. Ltd. (ZingHR)**

March 2023 – April 2024

Domain: Enterprise HRMS SaaS Platform (900+ Enterprise Clients)

- **Core Payroll Modernization:** Architected and led the engineering transformation of the core Payroll Engine into a highly generic, multi-tenant system to seamlessly accommodate diverse enterprise business rules across 900+ clients.

- **Microservices & Event Streaming:** Implemented scalable microservices architecture utilizing .NET 8, Apache Kafka for decoupled event messaging, and PostgreSQL for resilient high-volume transaction processing.
- **Delivery & Engineering Leadership:** Led end-to-end development, code reviews, technical documentation, architectural grooming, and daily Scrum ceremonies across cross-functional engineering squads.
- **Stakeholder & Client Alignment:** Partnered closely with Implementation Specialists, Delivery Owners, and Product Managers to translate complex HRMS functional requirements into scalable, reusable architectural solutions.

Technology Lead | Infogain Software Engineering Business Technology

September 2019 – March 2023

Client: Costco Wholesale Corporation (Fortune 14 Global Retailer) – US & Canada

- **Global eCommerce Solution Leadership:** Led end-to-end solution design, technical architecture, and delivery of the mission-critical eCommerce Tire Platform for Costco Wholesale (Fortune 14), supporting multi-million users across US and Canada.
- **Tire Inventory Microservice Module:** Spearheaded the Tire Inventory module architecture, leveraging PostgreSQL and Redis caching alongside enterprise systems, achieving sub-second catalog and inventory search performance.
- **Technology Stack:** ASP.NET MVC 5, C#, SQL Server, PostgreSQL, Redis, Azure Cloud, jQuery, HTML5, BlazeMeter, SonarQube.
- **Quality, Accessibility & Compliance:** Enforced stringent coding standards, WCAG/ADA accessibility compliance, SonarQube quality gates, and automated security vulnerability checks.
- **Performance Engineering & Scalability:** Conducted comprehensive load and stress testing using BlazeMeter and JMeter, optimizing database execution plans, API throughput, and caching strategies for peak retail seasons (Black Friday/Cyber Monday).
- **Client Steering & Delivery Management:** Led technical design reviews and roadmap discussions with Costco senior leadership and onsite architects; managed release planning, production rollouts, and L3 on-call incident resolution.
- **Award & Recognition:** Conferred the prestigious "Star of the Month" award for outstanding technical delivery, architectural excellence, and engineering leadership on the Costco platform.

Senior Associate – Technical Lead | Sportz Interactive Pvt. Ltd.

October 2016 – September 2019

Clients: UEFA Champions League, Yahoo Cricket, IPL, European Rugby (EPCR), ESPN CricInfo

- **High-Concurrency Gaming Platforms:** Architected and engineered backend systems for Daily Fantasy Cricket, Football, and Rugby games, delivering real-time, low-latency experiences during major global sporting events.
- **Distributed Web API & Services:** Engineered high-performance Web APIs and Windows background services using .NET Core, ASP.NET, AWS cloud services, and PostgreSQL databases.
- **Real-time Simulation & Point Calculation Engines:** Architected live match data ingestion pipelines, automated game simulation engines, real-time leaderboard ranking calculations, and instant point distribution logic.
- **Integrations & API Contracts:** Led 3rd-party sports data feed integrations and authored standardized API documentation (Swagger/OpenAPI) for seamless web/mobile client communication.

Programmer Analyst | Sodexo SVC India Pvt. Ltd.

October 2014 – October 2016

Projects: Enterprise Payment System (EPS), Voucher SWOS, SVAR, Marketing Applications

- **Payment & Voucher Systems Architecture:** Developed and optimized core modules for Sodexo Enterprise Payment Systems (EPS) and Voucher management suites using ASP.NET MVC and SQL Server.
- **Database Architecture & Optimization:** Architected and maintained complex database objects, stored procedures, triggers, and views; optimized heavy transactional queries for high-speed processing.

- **Modular Framework Refactoring:** Drove R&D initiatives to refactor siloed application components into reusable, generic enterprise libraries, reducing redundant development efforts by 30%.

Software Developer | Tara Jewels Pvt. Ltd. **October 2012 – October 2014**

Projects: Online Retail Portal, Import-Export Management, Product Catalog System

- **Full-Stack Application Development:** Developed responsive web applications for global jewelry retail, inventory tracking, import-export operations, and automated reporting using ASP.NET MVC, C#, SQL Server, and jQuery.
- **Database Design & Optimization:** Wrote optimized SQL queries, stored procedures, and complex data extraction scripts for inventory auditing and business reporting.

Software Developer | Aim Global Solutions Pvt. Ltd. **June 2011 – October 2012**

Projects: RFID Vehicle & Document Tracking, Prison Management System, Defense Solutions

- **RFID & Smart Device Engineering:** Engineered RFID tracking applications, handheld smart device software (Windows CE), and backend database systems in ASP.NET and SQL Server.
- **Client Demos & Deployment:** Collaborated on requirement gathering with defense and enterprise stakeholders; conducted live demonstrations and on-site client deployments.

EDUCATION & KEY ACHIEVEMENTS

Bachelor of Science in Information Technology (B.Sc. IT)
Mumbai University (Usha Pravin Gandhi College of Management)
Score: 71.33% (First Class with Distinction) | 2011

Higher Secondary Certificate (H.S.C.)
Maharashtra State Board (Mithibai College) | 68.33% | 2008

Secondary School Certificate (S.S.C.)
Maharashtra State Board (G.P.P. High School) | 77.87% | 2006

Key Honors & Awards:

"Star of the Month" Award
Infogain (Costco Wholesale Project Delivery Excellence)